EXP.NO: 02

# TRASIENT ANALYSIS OF SERIES RLC CIRCUIT

## AIM:

To determine transient response of a series RLC circuit using Multisim Simulator.

## APPARATUS REQUIRED:

| S.No | Components | Specification | Quantity |
|---|---|---|---|
| 1. | CRO | 10MHZ | 1 |
| 2. | DCB, DIB | --- | 1 |
| 3. | Resistors | 10kΩ | |
| 4. | Bread board | --- | 1 |
| 5. | Connecting wires | --- | --- |
| 6. | PC with Multisim simulator | --- | --- |

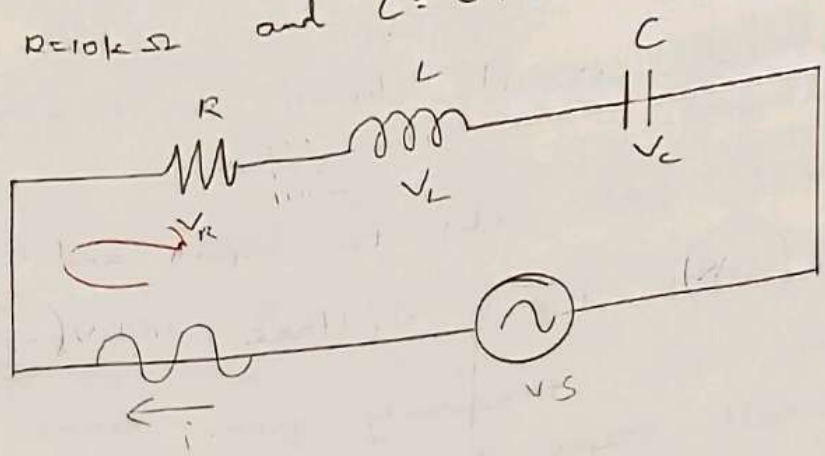
## PROCEDURE:

1. Make Connection as per the circuit diagram.
2. Calculate the value of $V_c(t)$ and wire in table. $V_R(t)$ and $V_L(t)$

Compare the calculated values and measured values of voltage in both the cases.

Circuit Diagram:-

Choose $R = 10k\Omega$ and $C = 0.1\mu F$



Tabulation:

Vin Amplitude: _____ Vin Frequence: _____ Time constant

RC = _____

| Time | $V_c(t)$ - charging cycle observed value | $V_c(t)$ - Discharging cycle observed value |
|---|---|---|
| $t=0$ | 0V | |
| $t=\tau$ | 2.5V | |
| $t=2\tau$ | 3.6V | |
| $t=3\tau$ | 4.3V | |
| $t=4\tau$ | 4.7V | |
| $t=5\tau$ | 5V | |

| time(t) | $V_L(t)$ - Voltage across inductor observed value |
|---|---|
| $t=0$ | 5V |
| $t=\tau$ | 3.0V |
| $t=2\tau$ | 1.8V |
| $t=3\tau$ | 1.0V |
| $t=4\tau$ | 0.5V |
| | |

| time(t) | $V_R(t)$ - Voltage across observed value |
|---|---|
| $t=0$ | 0V |
| $t=\tau$ | 2V |
| $t=2\tau$ | 3V |
| $t=3\tau$ | 4V |
| $t=4\tau$ | 4.5V |
| $t=5\tau$ | 5V |

| S.NO | DESCRIPTION | MARKS ALLOTED | MARKS OBTAINED |
|---|---|---|---|
| 1. | Experimental setup Connections | 20 | 20 |
| 2. | Observations | 20 | 18 |
| 3. | Calculations | 20 | 18 |
| 4. | Viva | 20 | 18 |
| 5. | Record | 20 | 20 |
| | | | 94 |

RESULT:

Thus the transient analysis of series RLC circuit is done practically and verified with theoretical value using multisim simulator.